Machine Learning Department
School of Computer Science
Carnegie Mellon University

PAC Learning

Matt Gormley
Lecture 15
Mar. 9, 2026

Q&A

Q: What is "bias"?

A: That depends. The word "bias" shows up all over machine learning! Watch out…

1. The additive term in a linear model (i.e. b in $w^T x + b$)
2. Inductive bias is the principle by which a learning algorithm generalizes to unseen examples
3. Bias of a model in a societal sense may refer to racial, socio-economic, gender biases that exist in the predictions of your model
4. The difference between the expected predictions of your model and the ground truth (as in "bias-variance tradeoff")

Reminders

- Homework 5: Neural Networks
  - Out: Wed, Feb 25
  - Due: Sun, Mar 15 at 11:59pm

LEARNING THEORY

PAC(-MAN) Learning

For some hypothesis $h \in \mathcal{H}$:

1. True Error

$$R(h)$$

2. Training Error

$$\hat{R}(h)$$

Question 2:

What is the expected number of PAC-MAN levels Matt will complete before a **Game-Over**?

A. 1-10
B. 11-20
C. 21-30

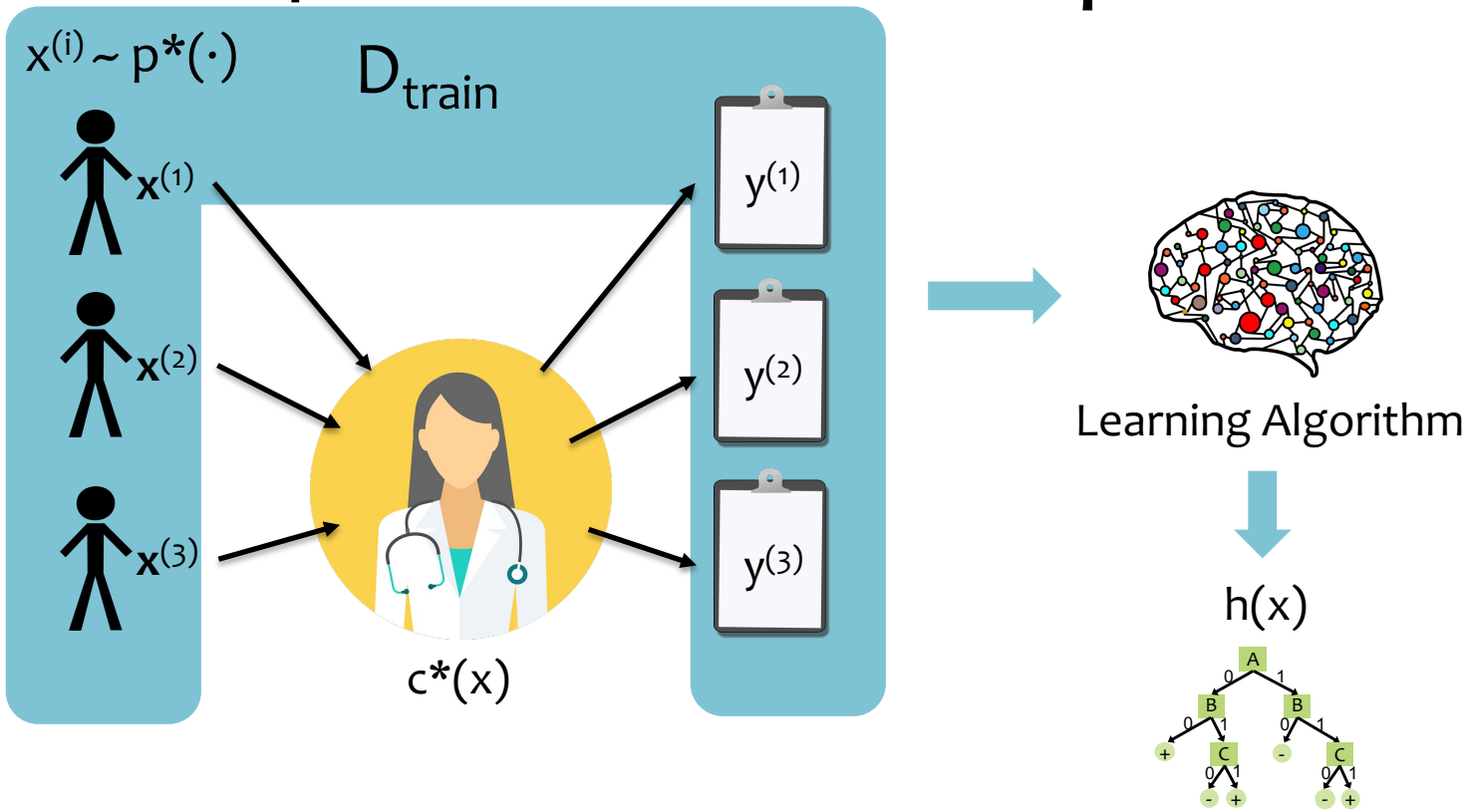
Questions for today (and next lecture)

1. Given a classifier with **zero training error**, what can we say about **true error** (aka. generalization error)?
   (Sample Complexity, Realizable Case)

2. Given a classifier with **low training error**, what can we say about **true error** (aka. generalization error)?
   (Sample Complexity, Agnostic Case)

3. Is there a **theoretical justification for regularization** to avoid overfitting?
   (Structural Risk Minimization)

PAC/SLT Model for Supervised ML



PAC/SLT Model for Supervised ML

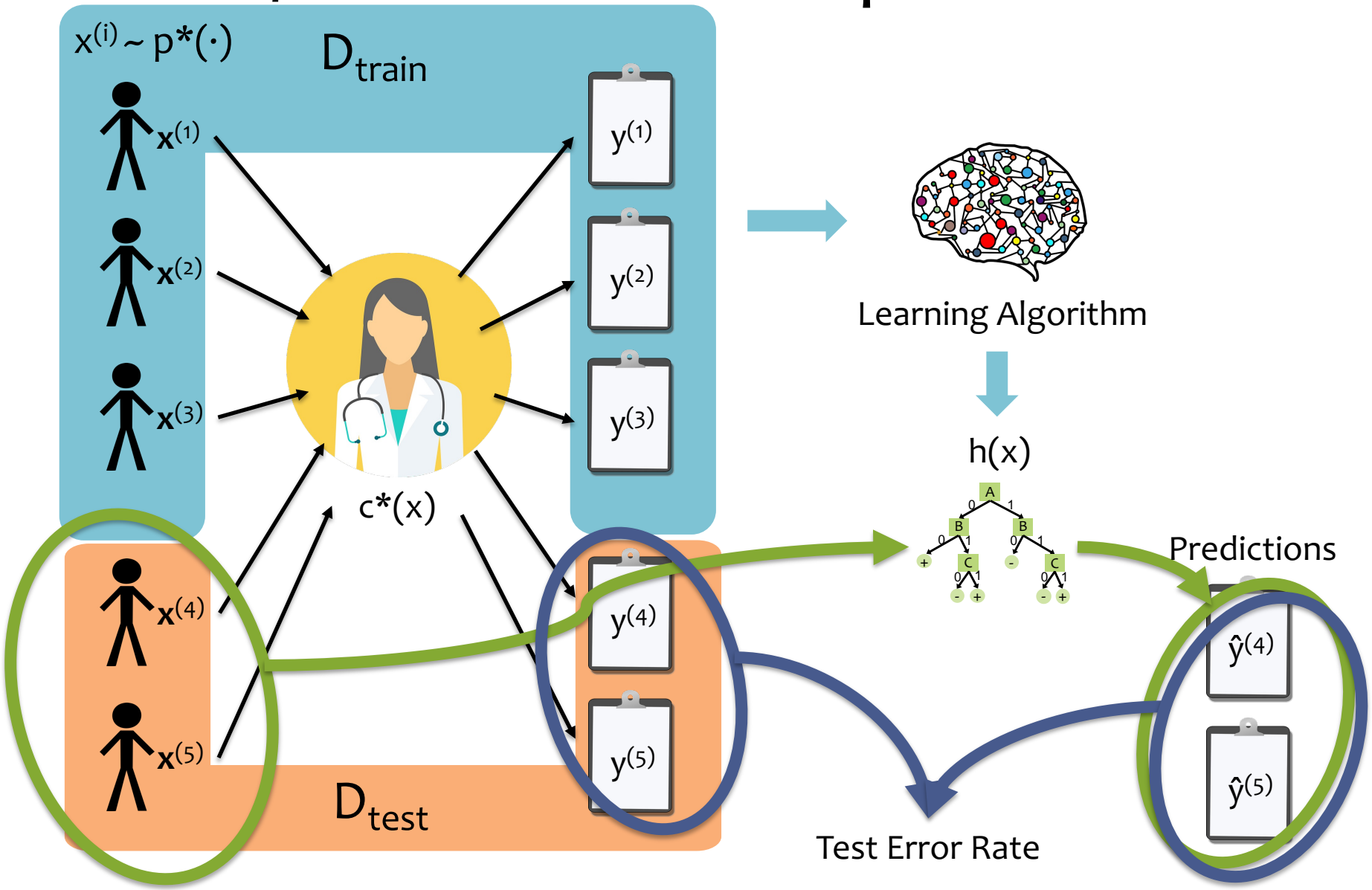
- **Problem Setting**
  - Set of possible inputs, $\mathbf{x} \in \mathcal{X}$ (all possible patients)
  - Set of possible outputs, $y \in \mathcal{Y}$ (all possible diagnoses)
  - Distribution over instances, $p^*(\cdot)$
  - Exists an unknown target function, $c^* : \mathcal{X} \rightarrow \mathcal{Y}$ (the doctor's brain)
  - Set, $\mathcal{H}$, of candidate hypothesis functions, $h : \mathcal{X} \rightarrow \mathcal{Y}$ (all possible decision trees)
- **Learner is given** N training examples $D = \{(\mathbf{x}^{(1)}, y^{(1)}), (\mathbf{x}^{(2)}, y^{(2)}), \ldots, (\mathbf{x}^{(N)}, y^{(N)})\}$ where $x^{(i)} \sim p^*(\cdot)$ and $y^{(i)} = c^*(\mathbf{x}^{(i)})$ (history of patients and their diagnoses)
- **Learner produces** a hypothesis function, $\hat{y} = h(x)$, that best approximates unknown target function $y = c^*(x)$ on the training data

IMPORTANT NOTE

In our discussion of PAC Learning, we are only concerned with the problem of **binary** classification

There are other theoretical frameworks (including PAC) that handle other learning settings, but this provides us with a representative one.

PAC/SLT Model for Supervised ML



Two Types of Error

1. True Error (aka. **expected risk**)

$$R(h) = P_{\mathbf{x} \sim p^*(\mathbf{x})}(c^*(\mathbf{x}) \neq h(\mathbf{x}))$$

This quantity is always **unknown**

2. Train Error (aka. **empirical risk**)

$$\begin{aligned}\hat{R}(h) &= P_{\mathbf{x} \sim \mathcal{S}}(c^*(\mathbf{x}) \neq h(\mathbf{x})) \\ &= \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(c^*(\mathbf{x}^{(i)}) \neq h(\mathbf{x}^{(i)})) \\ &= \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(y^{(i)} \neq h(\mathbf{x}^{(i)}))\end{aligned}$$

We can **measure** this on the training data

where $\mathcal{S} = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(N)})\}_{i=1}^{N}$ is the training data set, and $\mathbf{x} \sim \mathcal{S}$ denotes that $\mathbf{x}$ is sampled from the empirical distribution.

PAC / SLT Model

We've also referred to this as the "Function Approximation View"

1. Generate instances from *unknown* distribution p^*

$$\mathbf{x}^{(i)} \sim p^*(\mathbf{x}),\ \forall i \tag{1}$$

2. Oracle labels each instance with *unknown* function c^*

$$y^{(i)} = c^*(\mathbf{x}^{(i)}),\ \forall i \tag{2}$$

3. Learning algorithm chooses hypothesis $h \in \mathcal{H}$ with low(est) training error, $\hat{R}(h)$

$$\hat{h} = \operatorname*{argmin}_h \hat{R}(h) \tag{3}$$

4. Goal: Choose an h with low generalization error $R(h)$

Three Hypotheses of Interest

The **true function** c^* is the one we are trying to learn and that labeled the training data:

$$y^{(i)} = c^*(\mathbf{x}^{(i)}), \forall i \tag{1}$$

The **expected risk minimizer** has lowest true error:

$$h^* = \operatorname*{argmin}_{h \in \mathcal{H}} R(h) \tag{2}$$

The **empirical risk minimizer** has lowest training error:

$$\hat{h} = \operatorname*{argmin}_{h \in \mathcal{H}} \hat{R}(h) \tag{3}$$

ERM = the learning method that picks a hypothesis by minimizing empirical risk

Three Hypotheses of Interest

Poll Question 1: *True or False*: h* and c* are always equal.

$$y^{(i)} = c^*(\mathbf{x}^{(i)}), \forall i \qquad h^* = \underset{h \in \mathcal{H}}{\operatorname{argmin}}\, R(h)$$

Answer:

PAC Learning

- Q: Can we bound R(h) in terms of $\hat{R}$(h)?
- A: Yes!

- **PAC** stands for

Probably

Approximately

Correct

Why the name PAC?

A **PAC Learner** yields a hypothesis $h \in \mathcal{H}$ which is…

approximately correct $R(h) \approx 0$

with high probability $\Pr(R(h) \approx 0) \approx 1$

Probably Approximately Correct (PAC) Learning

1. PAC Criterion

Definition: The *PAC criterion* is that our learner produces a high accuracy learner with high probability:

$$P(|R(h) - \hat{R}(h)| \leq \epsilon) \geq 1 - \delta$$

- **Question:** What is random here?
- **Answer:**

2. Sample Complexity

Definition: The *sample complexity* is the minimum number of training examples N such that the PAC criterion is satisfied for some ϵ and δ.

3. Consistent Hypothesis

Definition: A hypothesis $h \in \mathcal{H}$ is *consistent* with the training data if

$$\hat{R}(h) = 0.$$

SAMPLE COMPLEXITY RESULTS (FOUR CASES)

Sample Complexity Results

Four Cases we care about…

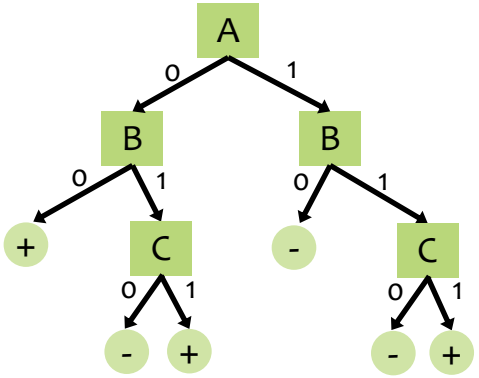
The size of the hypothesis space is a simple measure of model complexity

	Realizable: $c^* \in \mathcal{H}$	Agnostic: $c^* \in \mathcal{H}$ or $c^* \notin \mathcal{H}$
Finite $\lvert\mathcal{H}\rvert$: $\lvert\mathcal{H}\rvert < \infty$	We'll start with the finite, realizable case…	
Infinite $\lvert\mathcal{H}\rvert$: $\lvert\mathcal{H}\rvert = \infty$		

Finite vs. Infinite |H|

Finite |H|

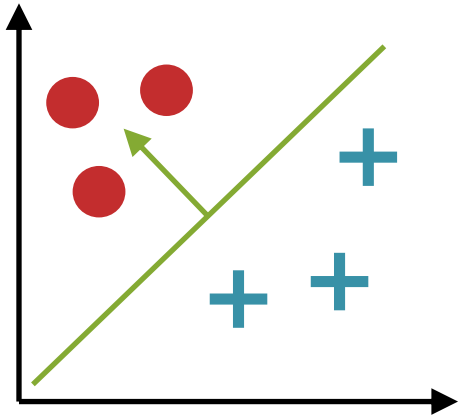
- *Example*: H = the set of all decision trees of depth D over binary feature vectors of length M



- *Example*: H = the set of all conjunctions over binary feature vectors of length M

Infinite |H|

- *Example*: H = the set of all linear decision boundaries in M dimensions



- *Example*: H = the set of all neural networks with 1-hidden layer with length M inputs

Realizable vs. Agnostic

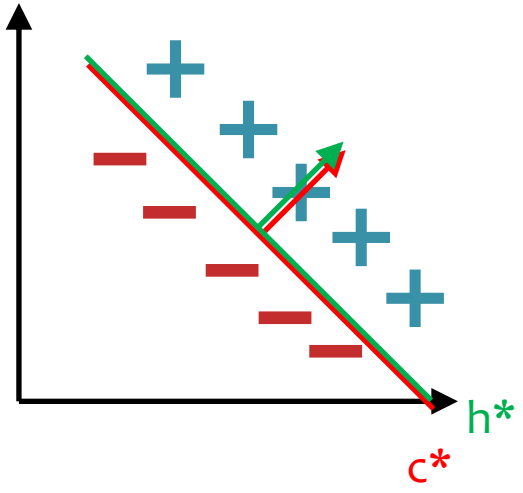
Realizable Case:
the true hypothesis is
definitely
in the hypothesis space

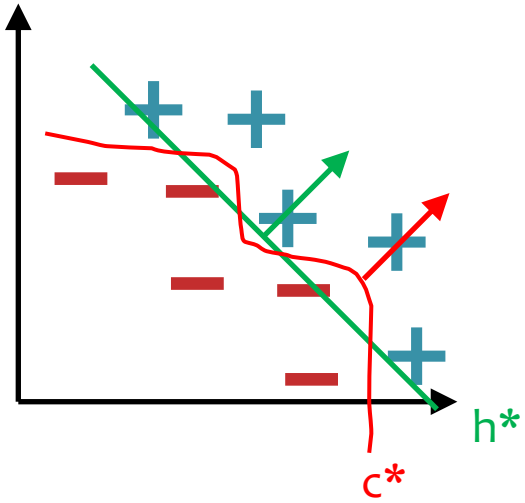
$$c^* \in \mathcal{H}$$

Agnostic Case:
the true hypothesis is
not necessarily
in the hypothesis space

$$c^* \in \mathcal{H} \text{ or}$$

$$c^* \notin \mathcal{H}$$





FINITE, REALIZABLE CASE

Theorem 1: Realizable Case, Finite |H|

Assume we have a learner which can find a hypothesis $h \in \mathcal{H}$ with zero training error $\hat{R}(h) = 0$.

Theorem 1: Finite, Realizable Case

For a finite hypothesis set $\mathcal{H}$ s.t. $c^* \in \mathcal{H}$ and arbitrary distribution p^*, if the number of labeled training data points satisfies

$$M \geq \frac{1}{\epsilon}\left(\ln\left(|\mathcal{H}|\right) + \ln\left(\frac{1}{\delta}\right)\right)$$

then with probability at least $1 - \delta$, all $h \in \mathcal{H}$ with $\hat{R}(h) = 0$ have

$$R(h) \leq \epsilon.$$

Example: Conjunctions

Poll Question 2:

Suppose H = class of conjunctions over **x** in $\{0,1\}^M$

Example hypotheses:

$h(\mathbf{x}) = x_1 (1-x_3) x_5$

$h(\mathbf{x}) = x_1 (1-x_2) x_4 (1-x_5)$

If M = 10, ε = 0.1, δ = 0.01, how many examples suffice according to Theorem 1?

Answer:

A. 10*(2*ln(10)+ln(100)) ≈ 92
B. 10*(3*ln(10)+ln(100)) ≈ 116
C. 10*(10*ln(2)+ln(100)) ≈ 116
D. 10*(10*ln(3)+ln(100)) ≈ 156
E. 100*(2*ln(10)+ln(10)) ≈ 691
F. 100*(3*ln(10)+ln(10)) ≈ 922
G. 100*(10*ln(2)+ln(10)) ≈ 924
H. 100*(10*ln(3)+ln(10)) ≈ 1329

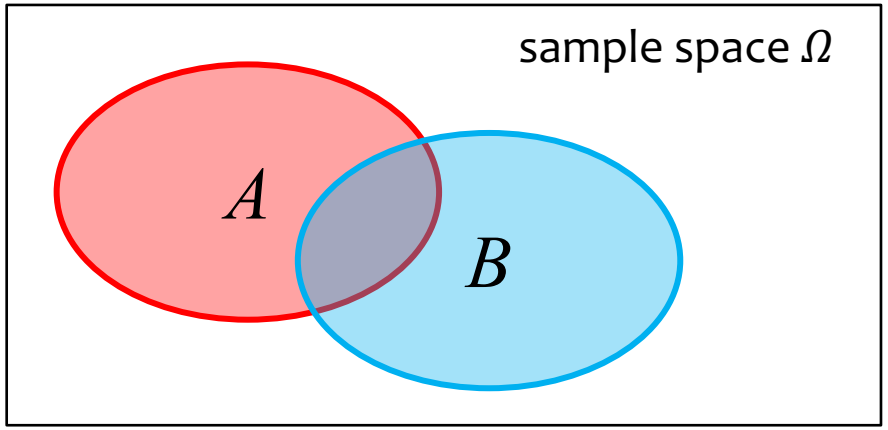
Thm. 1 $N \geq \frac{1}{\epsilon}\left[\log(|\mathcal{H}|) + \log(\frac{1}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ all $h \in \mathcal{H}$ with $\hat{R}(h) = 0$ have $R(h) \leq \epsilon$.

Background: Union Bound

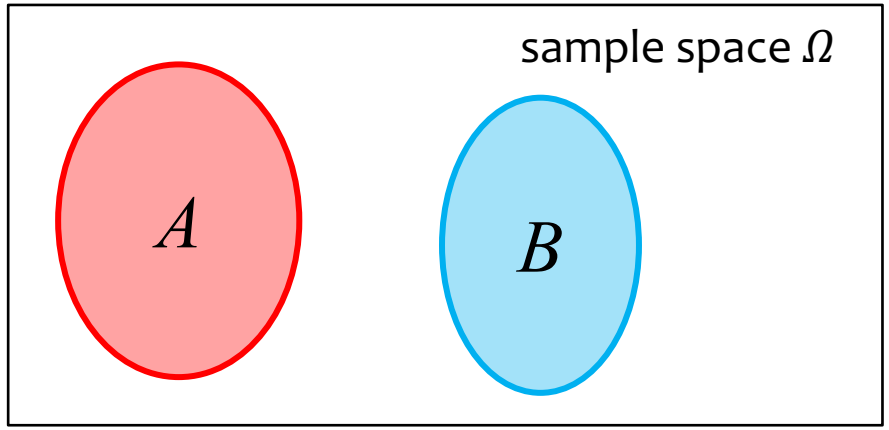
The **union bound** states that for two events A and B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$\leq P(A) + P(B)$$

Example 1:



Example 2:



Background: Contrapositive

- *Definition:* The **contrapositive** of the statement
$$A \Rightarrow B$$
is the statement
$$\neg B \Rightarrow \neg A$$
and the two are logically equivalent (i.e. they share all the same truth values in a truth table!)

- *Proof by contrapositive:*
If you want to prove $A \Rightarrow B$, instead prove $\neg B \Rightarrow \neg A$ and then conclude that $A \Rightarrow B$

- *Caution:* sometimes negating a statement is easier said than done, just be careful!

Proof of Theorem 1

Proof of Theorem 1

Proof of Theorem 1

FINITE, AGNOSTIC CASE

Theorem 1: Realizable Case, Finite |H|

Assume we have a learner which can find a hypothesis $h \in \mathcal{H}$ with zero training error $\hat{R}(h) = 0$.

Theorem 1: Finite, Realizable Case

For a finite hypothesis set $\mathcal{H}$ s.t. $c^* \in \mathcal{H}$ and arbitrary distribution p^*, if the number of labeled training data points satisfies

$$M \geq \frac{1}{\epsilon}\left(\ln\left(|\mathcal{H}|\right) + \ln\left(\frac{1}{\delta}\right)\right)$$

then with probability at least $1 - \delta$, all $h \in \mathcal{H}$ with $\hat{R}(h) = 0$ have

$$R(h) \leq \epsilon.$$

Theorem 2: Agnostic Case, Finite |H|

In **Empirical Risk Minimization** the learner finds the hypothesis $h \in \mathcal{H}$ with the smallest training error: $\hat{h} = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}(h)$.

Theorem 2: Finite, Agnostic Case

For a finite hypothesis set $\mathcal{H}$ and arbitrary distribution p^*, if the number of labelled training data points satisfies

$$M \geq \frac{1}{2\epsilon^2}\left(\ln\left(|\mathcal{H}|\right) + \ln\left(\frac{2}{\delta}\right)\right)$$

then with probability at least $1 - \delta$, all $h \in \mathcal{H}$ satisfy

$$\left|R(h) - \hat{R}(h)\right| \leq \epsilon$$

Sample Complexity Results

Definition 0.1. The **sample complexity** of a learning algorithm is the number of examples required to achieve arbitrarily small error (with respect to the optimal hypothesis) with high probability (i.e. close to 1).

Four Cases we care about…

	Realizable	Agnostic
Finite $\lvert\mathcal{H}\rvert$	**Thm. 1** $N \geq \frac{1}{\epsilon}\left[\log(\lvert\mathcal{H}\rvert) + \log(\frac{1}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ all $h \in \mathcal{H}$ with $\hat{R}(h) = 0$ have $R(h) \leq \epsilon$.	**Thm. 2** $N \geq \frac{1}{2\epsilon^2}\left[\log(\lvert\mathcal{H}\rvert) + \log(\frac{2}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ for all $h \in \mathcal{H}$ we have that $\lvert R(h) - \hat{R}(h)\rvert \leq \epsilon$.
Infinite $\lvert\mathcal{H}\rvert$		

1. Bound is **inversely linear in epsilon** (e.g. halving the error requires double the examples)
2. Bound is **only logarithmic in |H|** (e.g. squaring the size of the hypothesis space only doubles the log(|H|) term)

1. Bound is **inversely quadratic in epsilon** (e.g. halving the error requires 4x the examples)
2. Bound is **only logarithmic in |H|** (i.e. same as Realizable case)

	Realizable	Agnostic
Finite $\lvert\mathcal{H}\rvert$	**Thm. 1** $N \geq \frac{1}{\epsilon}\left[\log(\lvert\mathcal{H}\rvert) + \log(\frac{1}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ all $h \in \mathcal{H}$ with $\hat{R}(h) = 0$ have $R(h) \leq \epsilon$.	**Thm. 2** $N \geq \frac{1}{2\epsilon^2}\left[\log(\lvert\mathcal{H}\rvert) + \log(\frac{2}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ for all $h \in \mathcal{H}$ we have that $\lvert R(h) - \hat{R}(h)\rvert \leq \epsilon$.
Infinite $\lvert\mathcal{H}\rvert$		

SLT-style Corollaries

Thm. 1 $N \geq \frac{1}{\epsilon}\left[\log(|\mathcal{H}|) + \log(\frac{1}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ all $h \in \mathcal{H}$ with $\hat{R}(h) = 0$ have $R(h) \leq \epsilon$.

Solve the inequality in Thm.1 for epsilon to obtain Corollary 1

Corollary 1 (Realizable, Finite $|\mathcal{H}|$). For some $\delta > 0$, with probability at least $(1-\delta)$, for any h in $\mathcal{H}$ consistent with the training data (i.e. $\hat{R}(h) = 0$),

$$R(h) \leq \frac{1}{N}\left[\ln(|\mathcal{H}|) + \ln\left(\frac{1}{\delta}\right)\right]$$

We can obtain similar corollaries for each of the theorems…

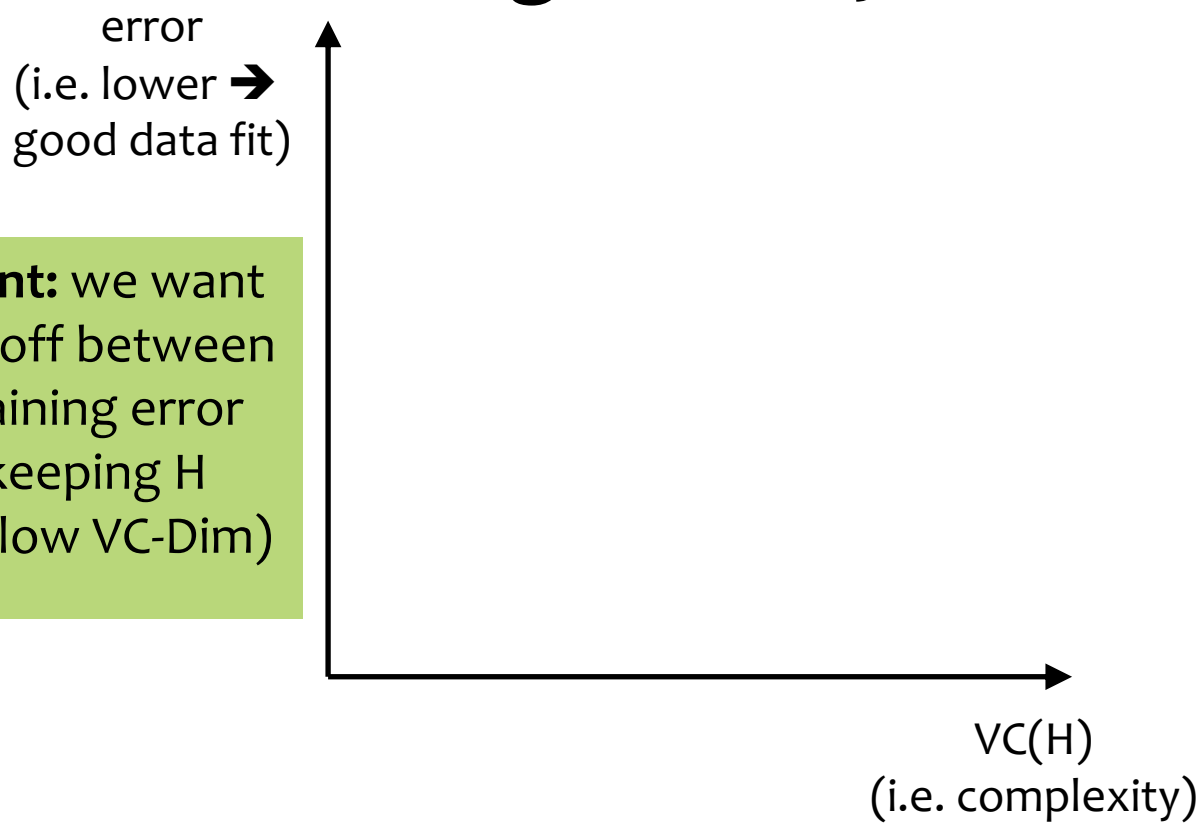
SLT-style Corollaries

Thm. 2 $N \geq \frac{1}{2\epsilon^2}\left[\log(|\mathcal{H}|) + \log(\frac{2}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ for all $h \in \mathcal{H}$ we have that $|R(h) - \hat{R}(h)| \leq \epsilon$.

Corollary 2 (Agnostic, Finite $|\mathcal{H}|$). For some $\delta > 0$, with probability at least $(1-\delta)$, for all hypotheses h in $\mathcal{H}$,

$$R(h) \leq \hat{R}(h) + \sqrt{\frac{1}{2N}\left[\ln(|\mathcal{H}|) + \ln\left(\frac{2}{\delta}\right)\right]}$$

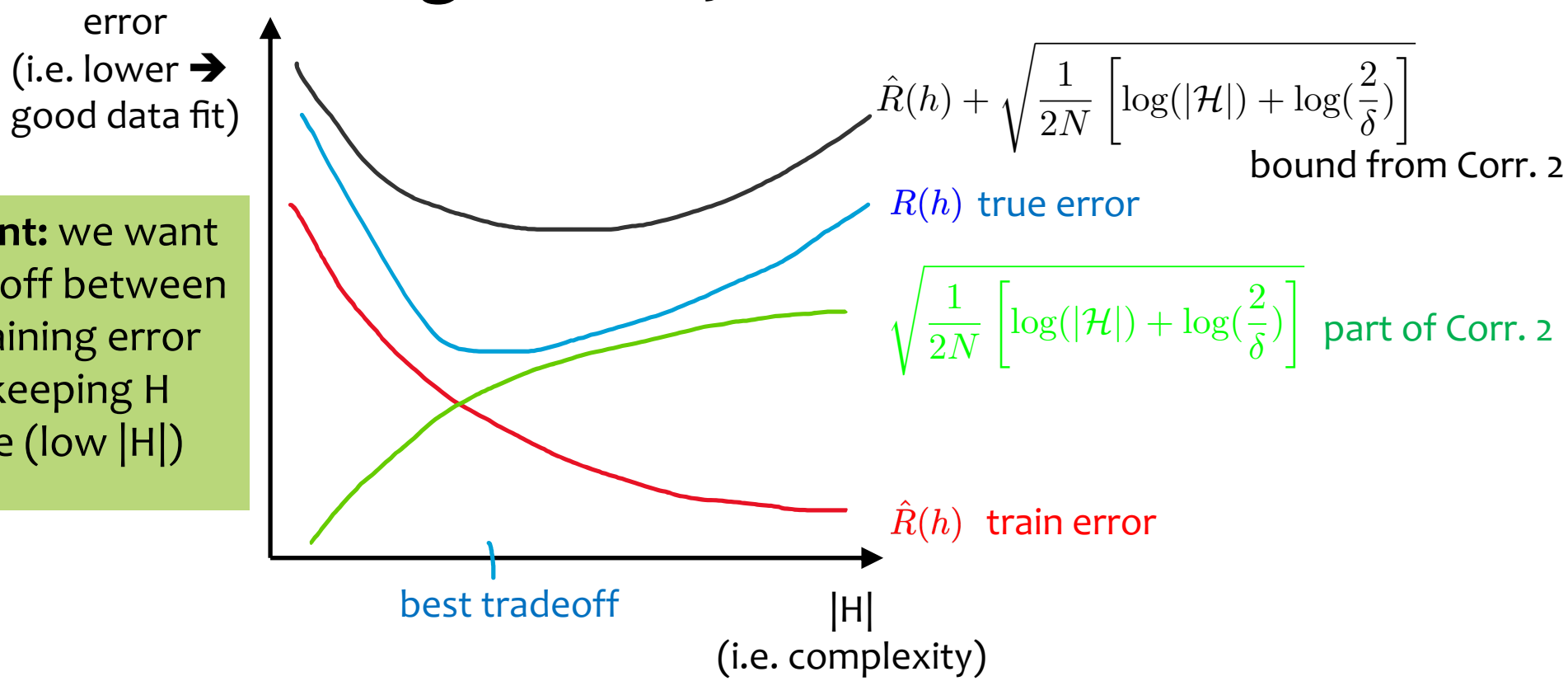
Learning Theory & Model Selection



Key Point: we want to tradeoff between low training error and keeping H simple (low VC-Dim)

Learning Theory & Model Selection

Key Point: we want to tradeoff between low training error and keeping H simple (low |H|)



Ex: H = conjunctions on d binary attributes:
ln(|H|) = d ln(3)
Expert sorts attributes, most likely to be relevant first

We allow conjunctions on first of them:

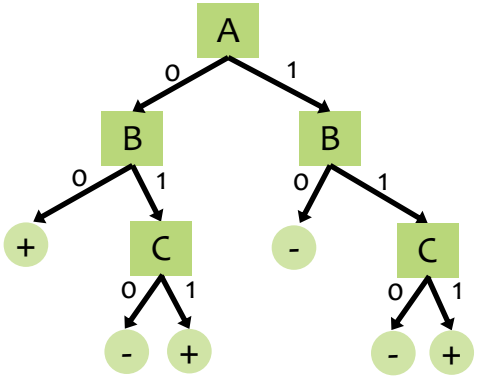
- training error goes down as d increases
- regularizer goes up as d increases
- stop when PAC bound is smallest (best tradeoff)

DON’T MOST OF OUR MODELS FALL UNDER THE INFINITE CASE?

Finite vs. Infinite |H|

Finite |H|

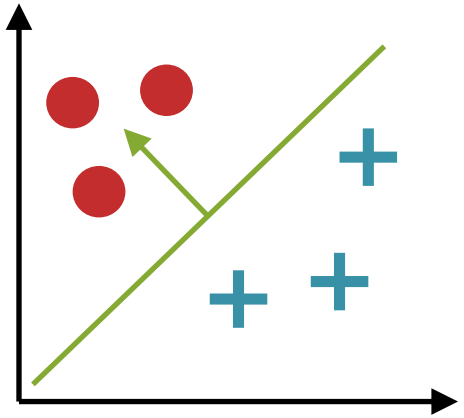
- *Example*: H = the set of all decision trees of depth D over binary feature vectors of length M



- *Example*: H = the set of all conjunctions over binary feature vectors of length M

Infinite |H|

- *Example*: H = the set of all linear decision boundaries in M dimensions



- *Example*: H = the set of all neural networks with 1-hidden layer with length M inputs

Sample Complexity Results

Definition 0.1. The **sample complexity** of a learning algorithm is the number of examples required to achieve arbitrarily small error (with respect to the optimal hypothesis) with high probability (i.e. close to 1).

Four Cases we care about…

	Realizable	Agnostic
Finite $\lvert\mathcal{H}\rvert$	**Thm** bele prob have $R(h) \leq \epsilon$.	have that $\lvert R(h) - \hat{R}(h)\rvert \leq \epsilon$.
Infinite $\lvert\mathcal{H}\rvert$		

For these two cases, we will use a new definition for the "complexity" of a Hypothesis space called **VC Dimension**

Sample Complexity Results

Definition 0.1. The **sample complexity** of a learning algorithm is the number of examples required to achieve arbitrarily small error (with respect to the optimal hypothesis) with high probability (i.e. close to 1).

Four Cases we care about…

	Realizable	Agnostic
Finite $\lvert\mathcal{H}\rvert$	**Thm. 1** $N \geq \frac{1}{\epsilon}\left[\log(\lvert\mathcal{H}\rvert) + \log(\frac{1}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ all $h \in \mathcal{H}$ with $\hat{R}(h) = 0$ have $R(h) \leq \epsilon$.	**Thm. 2** $N \geq \frac{1}{2\epsilon^2}\left[\log(\lvert\mathcal{H}\rvert) + \log(\frac{2}{\delta})\right]$ labeled examples are sufficient so that with probability $(1-\delta)$ for all $h \in \mathcal{H}$ we have that $\lvert R(h) - \hat{R}(h)\rvert \leq \epsilon$.
Infinite $\lvert\mathcal{H}\rvert$	**Thm. 3** N=$O(\frac{1}{\epsilon}\left[\text{VC}(\mathcal{H})\log(\frac{1}{\epsilon}) + \log(\frac{1}{\delta})\right])$ labeled examples are sufficient so that with probability $(1-\delta)$ all $h \in \mathcal{H}$ with $\hat{R}(h) = 0$ have $R(h) \leq \epsilon$.	**Thm. 4** N = $O(\frac{1}{\epsilon^2}\left[\text{VC}(\mathcal{H}) + \log(\frac{1}{\delta})\right])$ labeled examples are sufficient so that with probability $(1-\delta)$ for all $h \in \mathcal{H}$ we have that $\lvert R(h) - \hat{R}(h)\rvert \leq \epsilon$.